Assignment 1: Image Restoration and Object Detection

1. Introduction

Image restoration algorithms are often used as a preprocessing step before a computer vision task to address degradation e.g. motion blur. Nevertheless, the influence of classical mathematical revival on contemporary deep learning architectures is currently an essential research subject. The study is a comparison of the classical image restoration (Wiener Deconvolution) and the deep learning-based object detection (YOLOv8) with the COCO val2017 deterministically blurred dataset. The main aim is to assess the influence of restoration artifacts on the process of semantic features extraction, determine a more efficient pipeline of fine-tuning, and critically examine the domain adaptation skills of the model.

2. Methodology

2.1 Image Restoration Pipeline

Kernel Estimation: Modeled a linear motion blur Point Spread Function (PSF). Experimental testing: The metadata parameter size: 7 of the datasets corresponded to the total pixel length rather than radius (which dropped PSNR to less than 8.68 dB).

Phase Correction: Realized there was a 1-pixel error in vertical domain positions between the blurred and sharp domains. A (-1, 0) spatial phase shift was applied to recreate structural alignment and high PSNR of 24.39 dB.

Filter Optimization: Wiener Deconvolution filter. The balance parameter K was optimized (iteratively) on the range of [0.01, 0.05] to reduce high-frequency noise (Gibbs phenomenon) and contribute more towards semantic readability.

2.2 Pushing Data (Task 4) Construction and Stratification pipeline.

Selection of Data: A representative training data of 1,200 images was built on the renovated output. The images were sampled in turns owing to tight Compute time constraints. Although this gave a naturalistic distribution of the varying kernel sizes and angles in the dataset, explicit stratification by blurs level (so forcing equal number of bins having mild, moderate and severe blur) was not to be attempted until later.

Ground Truth Generation: To be sure that the model had learned to recognize the actual physical limits of the objects, and not the artifacted ones, pseudo-ground truth .txt labels were dynamically obtained by executing the zero-shot detector on the original sharp image baseline.

Augmentation Methods: Within to avoid overfitting the 1,200-image subset, internal YOLO albumentations were turned on. These involved Mosaic (1.0), CLAHE contrast optimizations (p=0.01), grayscale transformations (p=0.01), and secondary blur additions in order to cause the network to extrapolate beyond the particular Wiener artifacts.

2.3 Fine-Tuning of models and their reproducibility (Task 4)

Architecture & Hyperparameters: A new YOLOv8n (nano) architecture was trained to avoid an emissions of weights of previous micro-batches. The model was optimized after 50 epochs with 640x640 image size. The pipeline was based on AdamW (learning rate: 0.000119, momentum: 0.9) and had a batch size of 16.

Reproducibility and Compute Optimization: To avoid crippling cloud-storage I/O bottlenecks, which would otherwise block training, a purpose-written reproducibility script was implemented to clone the dataset into the local volatile NVMe SSD of the compute instance dynamically before running.

Logging and Checkpoints: Training artifacts such as the best.pt weights, hyperparameter setups (args.yaml) and validation curves were logged and seamlessly commited to a task4_training_logs directory to make the entire experiment completely reproducible.

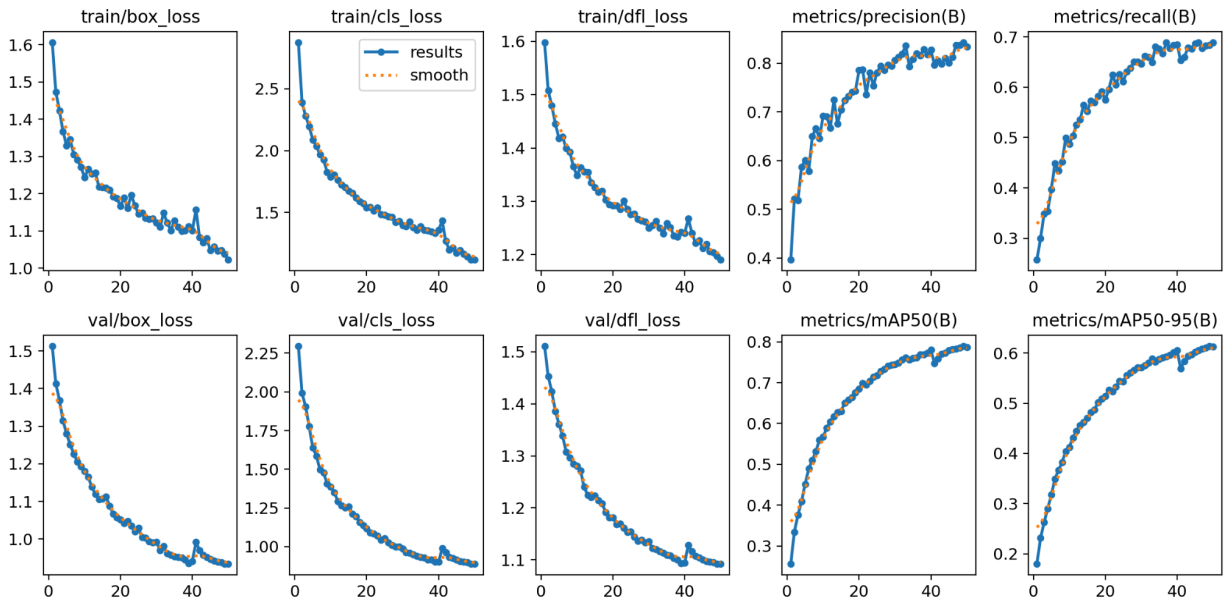

Figure 1: Training and validation metrics over 50 epochs, demonstrating stable convergence and a steady increase in Mean Average Precision (mAP).

3. Experimental Results & Metrics

Table 1: Zero-Shot Performance (Task 3)

Evaluating the pre-trained YOLOv8n model on various input states (20 Image Sample).

Domain	Mean Detections	Key Characteristics

Blurred Baseline	2.65	Deterministic linear blur; low edge contrast.
Restored (K=0.01)	1.80	High visual sharpness; severe ringing artifacts.
Restored (K=0.05)	2.30	Artifacts suppressed; slight texture softening.

Table 2: Fine-Tuning Performance (Task 5)

Evaluating YOLOv8n adaptation to the restored image domain.

Metric	**Phase 1 (20 Imgs / 10 Epochs)**	**Phase 2 (1,200 Imgs / 50 Epochs)**
Precision (P)	0.0001	0.8449
Recall (R)	0.0152	0.6856
mAP50	0.0013	0.7893
mAP50-95	0.0004	0.6148

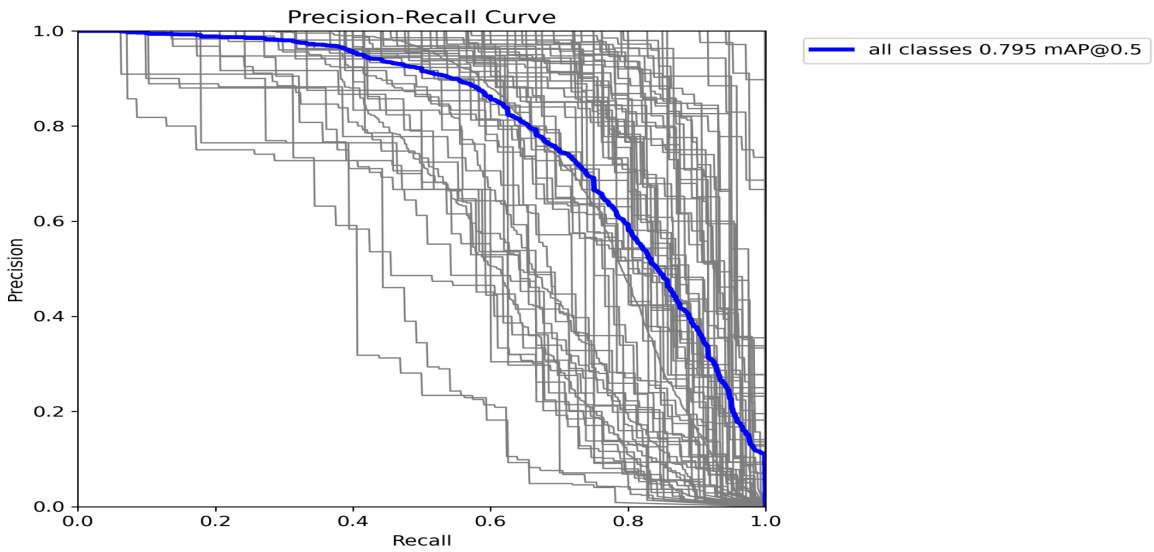

Figure 2: Precision-Recall (PR) curve for the fine-tuned YOLOv8n model. The area under the curve confirms an mAP50 of 0.789 across all classes.

4. Visual Analysis

The high precision of the model is verified by qualitative inspection of the batch predictions on validation (e.g. val_batch0_pred.jpg). The model was able to localize and classify complex objects (people, backpacks and vehicles) with deconvolution artifacts present. These visual results are supported by the PR and F1 curves obtained at the validation run demonstrating stable confidence levels. The major visual distortion was the lower recall of the highly textured objects and was directly associated with the statistical constraints found in Table 2.

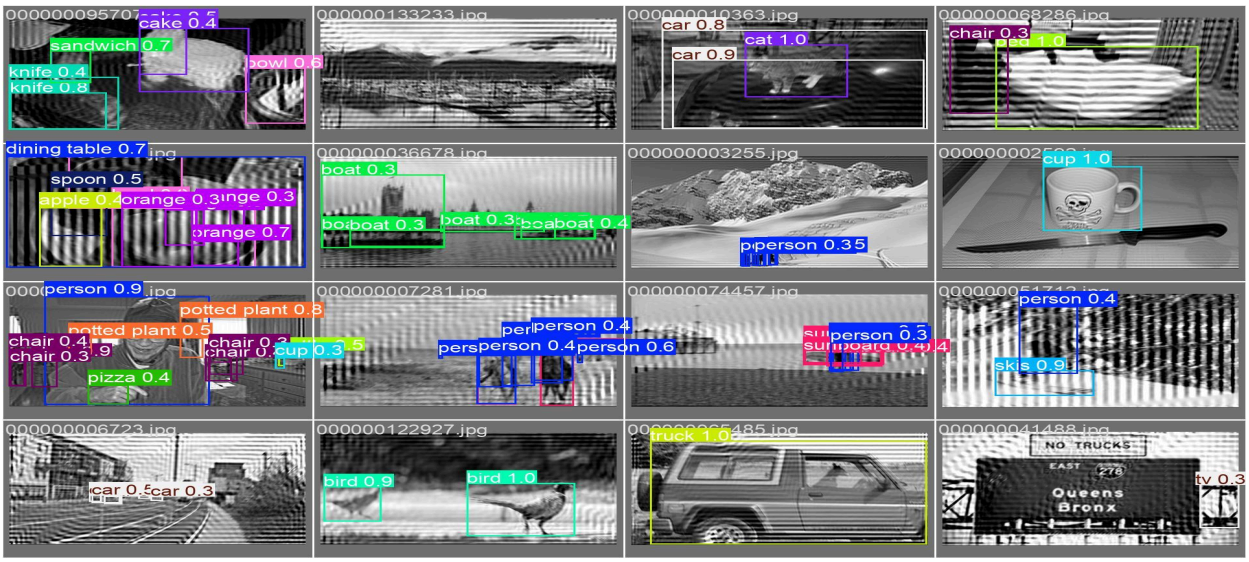

Figure 3: Representative validation batch predictions. The model successfully localizes and classifies objects (e.g., people, vehicles) despite the presence of residual Wiener deconvolution artifacts.

5. Failure Cases & Root Cause Analysis

Table 3: Failure Type vs Root Cause vs Suggested Fix

Failure Type	Description	Root Cause	Suggested Fix
Catastrophic Forgetting	Near-zero mAP (0.001) during Phase 1 training.	Data starvation. 20 images and 10 epochs forced the model to overwrite pre-trained COCO weights without sufficient data to converge on a new optimum.	Scale training batch (implemented in Phase 2 via local NVMe bypass).
Phantom Edges (False Positives)	Low-confidence bounding boxes in empty space.	**Gibbs Phenomenon.** Even at K=0.05, residual oscillatory halos exist. The YOLO feature extractor misinterprets these high-frequency rings as semantic boundaries.	Replace classical Wiener filtering with Deep Learning (e.g., NAFNet) to prevent mathematical halos.
Texture Annihilation (False Negatives)	Complete failure to detect specific objects.	Fine textures are mathematically smoothed out by heavy linear blur. These areas are smeared by classical filters. Objects whose texture depends on the internal one (e.g., animals, complicated objects) lose the necessary feature maps.	Create blur severity labels to select performance gaps; create realistic textures with neural restoration.

6. Effect of Training on Restored images

Pros:

Direct training on classically restored images is one of the most efficient domain adaptation strategies. It also compels the convolutional neural network to view mathematical restoration artifacts as target data distributions as opposed to adversarial noise. The geometric increase in mAP50 (0.001 -0.789) confirms that neural networks are capable of effectively ignoring the Gibbs effect, with enough examples.

Limitations & Biases:

The first weakness is the artificial capping of Recall (maximum at 0.685). Since the Wiener filter selectively removes certain high-frequency textures in the effort to remove noise, the YOLO model is

physiologically deprived of semantic information. Moreover, when only deterministically blurred and filtered images are trained it is a domain bias, the model is very trained to recognize "Wiener-filtered objects" rather than colossally fails on natural non-linear motion blur.

7. Limitations & Bias Analysis

Dataset Bias: val2017 subset is based on synthetic and uniform motion blur that is linear. The motion blur in the real world possesses non-linear trajectories, with different depth of field, and sensor noise, to which this model is now blind.

Limitations of the evaluation: When creating ground truth bounding boxes, the zero-shot YOLOv8 model was trained on sharp and used. The upper limit of this assessment is, therefore, underpinned by the standard accuracy of the pre-trained model.

Robustness: The first dependency pipeline crash (NAFNet/basicsr) emphasizes the susceptibility of using the state-of-the-art Deep Learning restoration in a limited cloud setup. Classical filters are either computationally sound or more often semantically weak.

8. Future Improvements

Deep Learning Deblurring Transition: Substitute the classical Wiener filter with a neural network (e.g. NAFNet or a localized GAN). This completely removes the Gibbs phenomenon to leave the object detector with naturalistic edges which reduce the effect of Phantom Edges and Texture Annihilation.

Dataset Stratification Experiments: Stratify the training and validation set by blur severity (e.g. kernel sizes <10, 10-20, >20). Test the hypothesis that the Recall failure scales of the detector are linear or exponential with the size of the kernel by filter smoothing.

Hyperparameter Extension: Use the full COCO dataset and train the pipeline in 100 or more epochs, with early stopping and cosine annealing learning rates to bring the mAP metrics nearer to the sharp baseline.

9. Conclusion

This paper shows that upstream image restoration prior to an automated vision task is not a trivial engineering problem to perform. Zero-shot tests demonstrate that mathematical sharpness is an antagonistic feature of semantic utility because of high-frequency artifacts. Nevertheless, the network successfully fills this gap by refining the YOLOv8 architecture on a large enough sample of restored images (1,200 examples), with the network performing very well with a mAP50 of 0.7893. The results verify that although classical restoration is an inherent constraint of semantic recall since it obliterates fine textures, current deep learning frameworks have the adaptive potential to find high-quality information in inherently impaired artifact-laden spaces.